

# Linearity of Expectation

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# Overview

## Linearity of Expectation (18.5)

- Expected Value of Two Dice (18.5.1)

- Sums of Indicator Random Variables (18.5.2)

- Expectation of a Binomial Distribution (18.5.3)

- The Coupon Collector Problem (18.5.4)

## Bayes Rule

## Linearity of expectation theorem

**Theorem:** For any random variables  $R_1, \dots, R_k$  and constants  $a_1, \dots, a_k \in \mathbb{R}$ ,

$$\mathbb{E} \left[ \sum_{i=1}^k a_i R_i \right] = \sum_{i=1}^k a_i \mathbb{E}[R_i].$$

**Proof:**

$$\begin{aligned} \mathbb{E} \left[ \sum_{i=1}^k a_i R_i \right] &= \sum_{\omega \in \mathcal{S}} \Pr(\omega) \left[ \sum_{i=1}^k a_i R_i(\omega) \right] && \text{defn expt} \\ &= \sum_{i=1}^k a_i \sum_{\omega \in \mathcal{S}} \Pr(\omega) [R_i(\omega)] && \text{move sum} \\ &= \sum_{i=1}^k a_i \mathbb{E}[R_i]. && \text{defn expt} \end{aligned}$$

# C'mon seven...

Let  $R_1$  and  $R_2$  be random variables corresponding to dice. What's the expected value of their sum? Outcome space is 36 different possible rolls. Linearity of expectation lets us short that out...

$$\mathbb{E}[R_1 + R_2] = \mathbb{E}[R_1] + \mathbb{E}[R_2] = 3.5 + 3.5 = 7, \text{ woo!}$$

Note that this result does *not* depend on independence. It holds even if we make the second die dependent on the first die, as long as it doesn't interfere with the expectation of the second die.

$$R_2 = R_1, R_2 = 7 - R_1, R_2 = \text{rem}((R_1 - 1) \times 101, 6) + 1.$$

# Shuffle 1 to 5

Start with the numbers 1 through 5, in order. Then, choose a random permutation. Look to see how many numbers stayed in their original place.

Example:

| 1 | 2 | 3 | 4 | 5 |   |
|---|---|---|---|---|---|
| 4 | 1 | 5 | 3 | 2 | 0 |
| 4 | 5 | 3 | 1 | 2 | 1 |
| 1 | 3 | 4 | 2 | 5 | 2 |

What's the expected number of numbers that stayed in their original place?

## The hard way

Let  $F(i)$  be the number of ways a permutation of 5 numbers can be arranged so exactly  $i$  numbers are in their original place.

The expected number of numbers that stayed in their original place is  $\sum_{i=0}^5 i F(i)/5!$ .

- ▶ What's  $F(5)$ ? 1.
- ▶ What's  $F(4)$ ? 0.
- ▶ What's  $F(0)$ ?  $4 \times$  huh. Turns out this value is the number of “derangements” of 5 items. Gets very tricky.
- ▶ What's  $F(1)$ ? It's 5 times the number of derangements of 4 items...

I think you can get there. But, it's a hike.

# The easy way

Let  $I_i$  be an indicator random variable on whether the  $i$ th number is in the  $i$ th place.

The answer we are looking for is

$$\mathbb{E}[I_1 + I_2 + I_3 + I_4 + I_5] = \mathbb{E}[I_1] + \mathbb{E}[I_2] + \mathbb{E}[I_3] + \mathbb{E}[I_4] + \mathbb{E}[I_5].$$

$\mathbb{E}[I_i]$  is the probability that a random permutation on 5 numbers puts number  $i$  in position  $i$ . It's  $1/5$ . So, the answer to the problem is  $1/5 + 1/5 + 1/5 + 1/5 + 1/5 = 1$ .

## The Harl- $n$ shuffle

Take the numbers 1 to  $n$  and scramble them.

Again, expected number of digits in the scrambled version that are in their original place is

$$\mathbb{E}\left[\sum_{i=1}^n I_i\right] = \sum_{i=1}^n \mathbb{E}[I_i] = \sum_{i=1}^n \frac{1}{n} = 1.$$

Interesting, right? Doesn't depend on  $n$ . As  $n$  increases, more positions, so less likely a random number will be in its original position. But, as  $n$  increases, more *chances* for a number to be in its right position. These forces balance.

140 students hand in a quiz. We shuffle them and redistribute so everyone will grade one quiz. What's the expected number of students who get their own quiz back? One.



# Expected number of events

Generalization of this idea.

**Theorem:** Given a set of events, the expected number of these events that occur is the sum of their probabilities.

That's true even if the events are not independent or if their probabilities sum to more than one, or whatever.

## Coin flip example

What's the expected number of heads if we flip a coin with probability  $p$  of coming up heads  $n$  times?

We know a lot about coin flips and number of ways to get  $k$  heads. We *can* work through the algebra and use the definition of expectation to get an answer.

Linearity of expectation is so much more direct! The expected number of heads from each flip is  $p$ . So, the expected number of heads in  $n$  flips is  $np$ . Done!

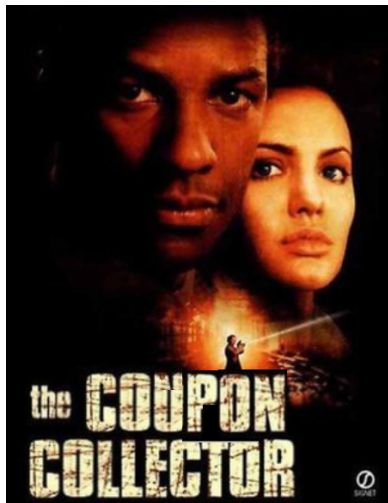
# Coupon collecting

Sometimes a company will run a coupon-collecting promotion. For example, each time you make a purchase, you get a coupon at random. Maybe the coupons are the letters of “PLANET.” If you collect all  $n = 6$  coupons, you win a large prize.

How many purchases should you expect to make to win (collect all  $n$  coupons) if coupon types are given uniformly at random at each purchase?

The problem is called the Coupon Collector problem, which sounds like a thriller to me.

## Movie poster



# Solving it

Consider a sequence of coupons that has all  $n$  types at the end. We want to know the *length* of this sequence.

Let  $X_i$  be the sequence of coupons from right *after* the  $(i - 1)$ th unique coupon was received until the  $i$ th unique coupon is received.

Examples:

$$\underbrace{3}_{X_1} \underbrace{4}_{X_2} \underbrace{5}_{X_3} \underbrace{3 \ 3}_{X_4} \underbrace{2 \ 2 \ 5 \ 1}_{X_5}$$

The length of the entire coupon sequence is the sum of the lengths of the  $X_i$ s:  $1 + 1 + 1 + 3 + 3 = 9$ .

The *expected* length of the entire sequence is the sum of the expected lengths of the  $X_i$ s.

# What's $X_k$ ?

What's the length of  $X_k$ ? At the beginning of segment  $k$ , we have collected  $k - 1$  different coupons from the  $n$  possibilities. Thus, the probability of getting one of the new ones is  $p = \frac{n-k+1}{n}$ .

What's the expected number of tries until a new one is found?  
 $\frac{n}{n-k+1}$  by the “mean time to failure” analysis!

|                                                 |                      |
|-------------------------------------------------|----------------------|
| $\mathbb{E}[\sum_{k=1}^n \text{length}(X_k)]$   | coup col             |
| $= \sum_{k=1}^n \mathbb{E}[\text{length}(X_k)]$ | lin of exp           |
| $= \sum_{k=1}^n n/(n - k + 1)$                  | mean fail time       |
| $= n \sum_{i=1}^n 1/i$                          | $i = n - k + 1$      |
| $= nH(n)$                                       | defn Harmonic number |
| $\approx n \ln(n)$                              | prop of $H$          |

# Examples

- ▶ People to poll before having representatives of each birthday:  
 $365 \times H(365) \approx 2365$
- ▶ Ice skates before you have one of each:  $2 \times H(2) = 3$
- ▶ Die rolls until you have one of each:  $6 \times H(6) = 14.7$
- ▶ Number of randomly assigned pigeons until each hole is filled:  
 $n \times H(n) \approx n \ln(n)$

## Flipping the conditional probability

**Theorem:**  $\Pr(A|B) = \Pr(B|A) \Pr(A) / \Pr(B)$ .

**Proof:**

|                   |                                    |                                  |
|-------------------|------------------------------------|----------------------------------|
| $\Pr(A B)$        | $= \Pr(A \text{ AND } B) / \Pr(B)$ | defn cond prob                   |
| $\Pr(A B) \Pr(B)$ | $= \Pr(A \text{ AND } B)$          | mult by $\Pr(B)$                 |
| $\Pr(B A)$        | $= \Pr(A \text{ AND } B) / \Pr(A)$ | defn cond prob                   |
| $\Pr(B A) \Pr(A)$ | $= \Pr(A \text{ AND } B)$          | mult by $\Pr(A)$                 |
| $\Pr(A B) \Pr(B)$ | $= \Pr(B A) \Pr(A)$                | both are $\Pr(A \text{ AND } B)$ |
| $\Pr(A B)$        | $= \Pr(B A) \Pr(A) / \Pr(B)$       | div by $\Pr(B)$                  |



## Virus test example

|        |                                |
|--------|--------------------------------|
| +test  | person tests positive          |
| +virus | person has the virus           |
| −test  | person tests negative          |
| −virus | person does not have the virus |

$$\Pr(+\text{test} \mid +\text{virus}) = 0.85 \quad \text{true positive}$$

$$\Pr(+\text{test} \mid -\text{virus}) = 0.10 \quad \text{false positive}$$

$$\Pr(+\text{virus}) = 0.01 \quad \text{population prob}$$

Test says you have the virus. What's the chance you have it?

$$\begin{aligned} &\Pr(+\text{virus} \mid +\text{test}) && \text{have it?} \\ &= \Pr(+\text{test} \mid +\text{virus}) \Pr(+\text{virus}) / \Pr(+\text{test}) && \text{Bayes rule} \\ &= 0.85 \times 0.01 / \Pr(+\text{test}) && \text{plug} \end{aligned}$$

$$\begin{aligned} &\Pr(-\text{virus} \mid +\text{test}) && \text{don't have it?} \\ &= \Pr(+\text{test} \mid -\text{virus}) \Pr(-\text{virus}) / \Pr(+\text{test}) && \text{Bayes rule} \\ &= 0.10 \times 0.99 / \Pr(+\text{test}) && \text{plug} \end{aligned}$$

## Probability of positive test

Either you have the virus, or you don't. So,

$$\begin{aligned} & \Pr(+\text{virus} \mid +\text{test}) + \Pr(-\text{virus} \mid +\text{test}) \\ &= 0.85 \times 0.01 / \Pr(+\text{test}) + 0.10 \times 0.99 / \Pr(+\text{test}) = 1 \end{aligned}$$

$$\Pr(+\text{test}) = 0.85 \times 0.01 + 0.10 \times 0.99 = 0.1075. \text{ So,}$$

$$\Pr(+\text{virus} \mid +\text{test}) = 0.85 \times 0.01 / 0.1075 \approx 8\%.$$

Positive test makes you much more likely to have the virus. But, it's still unlikely.